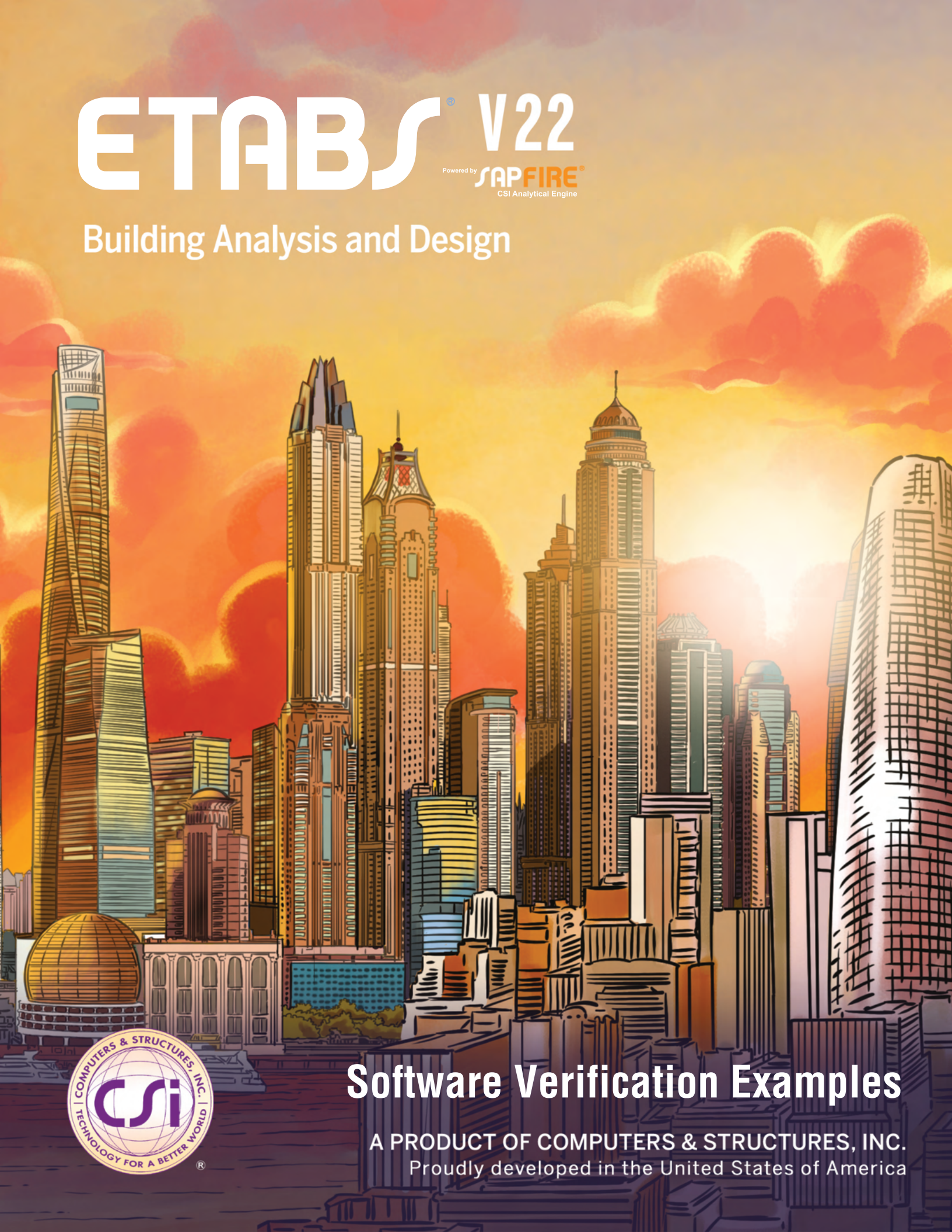
ETABS® V22

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Building Analysis and Design



Software Verification Examples

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Software Verification Examples

For

ETABS®

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AISC 360-22 Example 003 (BCf-S)	Beam to Column Flange - Shear Connection
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ACI 318-14 Example 001	Beam Shear & Flexural Reinforcing
ACI 318-14 Example 002	P-M Interaction Check for Rectangular Column
ACI 318-11 Example 001	Beam Shear & Flexural Reinforcing
ACI 318-11 Example 002	P-M Interaction Check for Rectangular Column
ACI 318-08 Example 001	Beam Shear & Flexural Reinforcing
ACI 318-08 Example 002	P-M Interaction Check for Rectangular Column
AS 3600-2009 Example 001	Beam Shear & Flexural Reinforcing

AS 3600-2009 Example 002	P-M Interaction Check for Rectangular Column
AS 3600-2018 Example 001	Beam Shear & Flexural Reinforcing
AS 3600-2018 Example 002	P-M Interaction Check for Rectangular Column
BS 8110-1997 Example 001	Beam Shear & Flexural Reinforcing
BS 8110-1997 Example 002	P-M Interaction Check for Rectangular Column
CSA A23.3-14 Example 001	Beam Shear & Flexural Reinforcing
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EN 2-2004 Example 001	Beam Shear & Flexural Reinforcing
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IS 456-2000 Example 001	Beam Shear & Flexural Reinforcing
IS 456-2000 Example 002	P-M Interaction Check for Rectangular Column
KBC 2009 Example 001	Beam Shear & Flexural Reinforcing
KBC 2009 Example 002	P-M Interaction Check for Rectangular Column
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KBC 2016 Example 002	P-M Interaction Check for Rectangular Column
KCI 1999 Example 001	Beam Moment Strength Using Equivalent Rectangular Stress Distribution
KCI 1999 Example 002	P-M Interaction Check for Rectangular Column
NTC 2008 Example 001	Beam Shear & Flexural Reinforcing

NTC 2008 Example 002	P-M Interaction Check for Rectangular Column
NTC 2008 Example 003	P-M Interaction Check for Compression Controlled Rectangular Column
NTC 2008 Example 004	P-M Interaction Check for Compression Controlled Rectangular Column
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SS CP 65-1999 Example 002	P-M Interaction Check for Rectangular Column
TS 500-2000 Example 001	Beam Shear & Flexural Reinforcing
TS 500-2000 Example 002	P-M Interaction Check for Rectangular Column
TS 500-2000(R2018) Example 001	Beam Shear & Flexural Reinforcing
TS 500-2000(R2018) Example 002	P-M Interaction Check for Rectangular Column

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ACI 318-14 WALL-001	P-M Interaction Check for Wall
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AS 3600-2018 WALL-001	P-M Interaction Check for a Wall
AS 3600-2018 WALL-002	P-M Interaction Check for a Wall
BS 8110-97 WALL-001	P-M Interaction Check for a Wall
BS 8110-97 WALL-002	P-M Interaction Check for a Wall
CSA A23.3-14 WALL-001	P-M Interaction Check for a Wall
CSA A23.3-14 WALL-002	P-M Interaction Check for a Wall
EC 2-2004 WALL-001	P-M Interaction Check for a Wall
EC 2-2004 WALL-002	P-M Interaction Check for a Wall
Hong Kong CP-2013 WALL-001	P-M Interaction Check for a Wall
Hong Kong CP-2013 WALL-002	P-M Interaction Check for a Wall
Indian IS 456-2000 WALL-001	P-M Interaction Check for a Wall
Indian IS 456-2000 WALL-002	P-M Interaction Check for a Wall
KBC 2009 WALL-001	P-M Interaction Check for Wall
KBC 2009 WALL-002	P-M Interaction Check for Wall
KBC 2016 WALL-001	P-M Interaction Check for Wall
KBC 2016 WALL-002	P-M Interaction Check for Wall
Mexican RCDF-04 WALL-001	P-M Interaction Check for a Wall
Mexican RCDF-04 WALL-002	P-M Interaction Check for a Wall

Mexican RCDF-2017 WALL-001	P-M Interaction Check for a Wall
Mexican RCDF-2017 WALL-002	P-M Interaction Check for a Wall
NZS-3103-2006 WALL-001	P-M Interaction Check for a Wall
NZS-3103-2006 WALL-002	P-M Interaction Check for a Wall
Singapore CP65-99-001	P-M Interaction Check for a Wall
Singapore CP65-99-002	P-M Interaction Check for a Wall
Turkish TS 500-2000 WALL-001	P-M Interaction Check for a Wall
Turkish TS 500-2000 WALL-002	P-M Interaction Check for a Wall
Turkish TS 500-2000(R2018) WALL-001	P-M Interaction Check for a Wall
Turkish TS 500-2000 WALL-002	P-M Interaction Check for a Wall

Composite Beam

AISC 360-05 Example 001	Composite Girder Design
AISC 360-10 Example 001	Composite Girder Design
AISC 360-10 Example 002	Composite Girder Design
AISC 360-16 Example 001	Composite Girder Design (Perpendicular Deck)
AISC 360-16 Example 002	Composite Girder Design (Parallel Deck)
AISC 360-16 Example 003	Composite Girder Design (w/ Axial Load)
AISC 360-22 Example 001	Composite Girder Design (Perpendicular Deck)
AISC 360-22 Example 002	Composite Girder Design (Parallel Deck)
AISC 360-22 Example 003	Composite Girder Design (w/ Axial Load)
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AISC DG02 Example 002	Steel Beam with Opening
AISC DG02 Example 003	Composite Beam with Opening
AISC DG02 Example 004	Composite Beam with Opening

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AISC DG31 Example 001	Non-composite Castellated Beam
AISC DG31 Example 002	Non-composite Cellular Beam
AISC DG31 Example 003	Composite Castellated Beam
AISC DG31 Example 004	Composite Cellular Beam
BS 5950-90 Example 001	Steel Designers Manual Sixth Edition – Design of Simply Supported Composite Beam
CSA S16-14 Example 001	Handbook of Steel Construction Tenth Edition – Composite Beam
CSA S16-19 Example 001	Handbook of Steel Construction Tenth Edition – Composite Beam
CSA S16-24 Example 001	Handbook of Steel Construction Tenth Edition – Composite Beam
EC 4-2004 Example 001	Steel Designers Manual Seventh Edition – Design of Simply Supported Composite Beam
IS-11384-2022 Example 001	INSDAG Teaching Materials – Composite Beam
SCI Publication 355 Example 001	Composite Beam with Web opening

Composite Column

AISC 360-10 Example 001	Composite Column Design
AISC 360-10 Example 002	Composite Column Design
AISC 360-10 Example 003	Composite Column Design
AISC 360-16 Example 001	Composite Column Design – Concrete Filled HSS Tube
AISC 360-16 Example 002	Composite Column Design– Concrete Filled HSS Tube
AISC 360-16 Example 003	Composite Column Design– Concrete Filled HSS Tube
AISC 360-16 Example 004	Composite Column Design – Concrete Encased Wide Flange

AISC 360-22 Example 001	Composite Column Design – Concrete Filled HSS Tube
AISC 360-22 Example 002	Composite Column Design– Concrete Filled HSS Tube
AISC 360-22 Example 003	Composite Column Design– Concrete Filled HSS Tube
AISC 360-22 Example 004	Composite Column Design – Concrete Encased Wide Flange
AISC 360-22 Example 005	Composite Column Design – Concrete Encased Wide Flange
AISC 360-22 Example 006	Composite Column Design – Concrete Encased Wide Flange
AISC 360-16 Example 005	Composite Column Design – Concrete Encased Wide Flange
AISC 360-16 Example 006	Composite Column Design – Concrete Encased Wide Flange
EC-4-2004 Example 001	Composite Column Design – Rectangular
EC-4-2004 Example 002	Composite Column Design- Concrete Filled CHS
IS11384-2022 Example 001	Composite Column Design – Rectangular
IS11384-2022 Example 002	Composite Column Design- Concrete Filled CHS
CSA-S16-19 Example 001	Composite Column Design – Concrete Filled Tube
CSA-S16-19 Example 002	Composite Column Design – Concrete Encased Wide Flange
CSA-S16-24 Example 001	Composite Column Design – Concrete Filled Tube
CSA-S16-24 Example 002	Composite Column Design – Concrete Encased Wide Flange

Slab

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ACI 318-19 RC-PN Ex001	Slab Punching Shear Design
ACI 318-19 RC-PN Ex002	Slab Punching Shear (Circular Column)
ACI 318-19 RC-SL Ex001	Slab Flexural Design

ACI 318-14 PT-SL Ex001	Post-Tensioned Slab Design
ACI 318-14 RC-PN Ex001	Slab Punching Shear Design
ACI 318-14 RC-SL Ex001	Slab Flexural Design
ACI 318-11 PT-SL Ex001	Post-Tensioned Slab Design
ACI 318-11 RC-PN Ex001	Slab Punching Shear Design
ACI 318-11 RC-SL Ex001	Slab Flexural Design
ACI 318-08 PT-SL Ex001	Post-Tensioned Slab Design
ACI 318-08 RC-PN Ex001	Slab Punching Shear Design
ACI 318-08 RC-SL Ex001	Slab Flexural Design
AS 3600-2009 PT-SL Ex001	Post-Tensioned Slab Design
AS 3600-2009 RC-PN Ex001	Slab Punching Shear Design
AS 3600-2009 RC-SL Ex001	Slab Flexural Design
BS 8110-1997 PT-SL Ex001	Post-Tensioned Slab Design
BS 8110-1997 RC-PN Ex001	Slab Punching Shear Design
BS 8110-1997 RC-SL Ex001	Slab Flexural Design
CSA A23.3-14 PT-SL Ex001	Post-Tensioned Slab Design
CSA A23.3-14 RC-PN Ex001	Slab Punching Shear Design
CSA A23.3-14 RC-SL Ex001	Slab Flexural Design
CSA A23.3-19 PT-SL Ex001	Post-Tensioned Slab Design
CSA A23.3-19 RC-PN Ex001	Slab Punching Shear Design
CSA A23.3-19 RC-SL Ex001	Slab Flexural Design
EN 2-2004 PT-SL Ex001	Post-Tensioned Slab Design
EN 2-2004 RC-PN Ex001	Slab Punching Shear Design
EN 2-2004 RC-SL Ex001	Slab Flexural Design
HK CP-2013 PT-SL Ex001	Post-Tensioned Slab Design
HK CP-2013 RC-PN Ex001	Slab Punching Shear Design
HK CP-2013 RC-SL Ex001	Slab Flexural Design

IS 456-2000 PT-SL Ex001	Post-Tensioned Slab Design
IS 456-2000 RC-PN Ex001	Slab Punching Shear Design
IS 456-2000 RC-SL Ex001	Slab Flexural Design
NTC 2008 PT-SL Ex001	Post-Tensioned Slab Design
NTC 2008 RC-PN Ex001	Slab Punching Shear Design
NTC 2008 RC-SL Ex001	Slab Flexural Design
NZS 3101-2006 PT-SL Ex001	Post-Tensioned Slab Design
NZS 3101-2006 RC-PN Ex001	Slab Punching Shear Design
NZS 3101-2006 RC-SL Ex001	Slab Flexural Design
SS CP 65-1999 PT-SL Ex001	Post-Tensioned Slab Design
SS CP 65-1999 RC-PN Ex001	Slab Punching Shear Design
SS CP 65-1999 RC-SL Ex001	Slab Flexural Design
TS 500-2000 PT-SL Ex001	Post-Tensioned Slab Design
TS 500-2000 RC-PN Ex001	Slab Punching Shear Design
TS 500-2000 RC-SL Ex001	Slab Flexural Design

LOG

ETABS Software Verification Log		
Revision Number	**Date**	**Description**
0	19 Apr 2013	Initial release of ETABS, Version 13.0.0
1	9 July 2013	• Minor documentation errors in the Verification manuals have been corrected • Minor improvements have been made to some of the examples, and some example file names have been changed for consistency. The design results produced and reported by ETABS are correct. The reported results are not changed except where the model has been changed. • Three new examples have been added for steel frame design.
2	11 Apr 2014	• Analysis model EX8.EDB - The response-spectrum function damping was incorrect and did not match the response-spectrum load case damping, hence the results produced did not match the documented value. After correction, the example produces the expected and documented results. No change was made to the Verification manual. • Analysis Example 03 - The name of code IBC2000 was changed to ASCE 7-02, as actually used in ETABS (IBC2000 was used in v9.7.4). In addition, the Verification manual was corrected for the actual values produced by ETABS. These values have not changed since v13.0.0. The documented values were for ETABS v9.7.4 and some changed in v13.0.0 due to the use of a different solver. The change has no engineering significance. • Analysis Example 06 and Example 07 - The Verification manual was corrected for the actual values produced by ETABS. These values have not changed since v13.0.0. The documented values were for ETABS v9.7.4 and some changed in v13.0.0 due to the use of a different solver. The change has no engineering significance. • Analysis Example 15 - The Verification manual was corrected for the actual values produced by ETABS. These values have not changed since v13.0.0. The documented values were for ETABS v9.7.4 and some changed in v13.0.0 due to the use of a different solver, and due to the difference in how wall

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		elements are connected to beams. The change due to the solver has no engineering significance. The change for wall elements was an enhancement. • Concrete Frame Design EN 2-2004 Example 001, Concrete Frame Design NTC 2008 Example 002 - The values produced by ETABS 2014 were updated in the Verification manual for a change in v13.1.3 under Incident 59154 (Ticket 23901) where the coefficients Alpha_CC and Alpha_LCC were not taken into account in certain cases. • Concrete Frame Design AS 3600-2009 Example 002, Shear Wall Design AS 3600-2009 WALL-002 - The values produced by ETABS were updated in the Verification manual for a change in v13.1.4 under Incident 59973 where the phi factor was incorrectly computed.
3	3 Nov 2014	• Analysis Example 14 – Minor changes have been made to the results as the result of an enhancement made under Incident 67283 to improve the convergence behavior of nonlinear static and nonlinear direct-integration time history analysis. • Composite Beam Design AISC-360-05 Example 001 was updated to reflect the fact that, under Incident 59912 it is now possible to specify that the shear stud strength is to be computed assuming the weak stud position. A typo in the version number of the referenced Design Guide example was corrected. A slight error in the hand-calculation for the partial composite action ΦM_n was corrected, resulting in perfect agreement with the value produced by ETABS. • Composite Beam Design AISC-360-10 Example 001 was updated to reflect the fact that, under Incident 59912 it is possible to specify that the shear stud strength is to be computed assuming the weak stud position. The hand-calculation for the partial composite action ΦM_n was revised to account for a lower percentage of composite action caused by an increase in the number of shear studs per deck rib in places, and a corresponding decrease in shear stud strength. • Composite Beam Design BS-5950-90 Example 001- The hand-calculations in the Verification manual were updated to reflect the actual section area of a UKB457x191x167, which differs from the value in the reference example, and to reflect that the maximum number of shear studs that can be placed on

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		the beam is 78 studs and not the 80 the reference example calls for. Also the value of the live load deflection produced by ETABS was updated for a change in v13.2.0 under Incident 56782. • Composite Beam Design CSA-S16-09 Example 001. The values produced by ETABS for the shear stud capacity were updated in the Verification manual for a change in v13.2.0 under Incident 71303. This change in turn affects the value of the partial composite moment capacity M_c but has no engineering significance. A typo affecting the value of pre-composite deflection in the Results Comparison table was corrected. • Composite Beam Design EC-4-2004 Example 001. The hand-calculation of the construction moment capacity, $M_{a,pl,Rd}$ was updated to reflect a more accurate value of the section W_{pl} and typos affecting the pre-composite deflection and beam camber were corrected. None of the values computed by ETABS changed.
4	7 Jan 2015	Initial release of ETABS 2015, Version 15.0.0 • Shear Wall Design example Eurocode 2-2004 Wall-002 has been updated due to changes previously reported under Incident #56569. • Shear Wall Design example AS 3600-09 Wall-001 has been updated due to changes previously reported under Incident #56113. • Shear Wall Design example CSA A23.3-04 Example 001 has been updated due to changes previously reported under Incident #71922. • Concrete Frame Design example CSA A23.3-04 Example 002 has been updated due to changes previously reported under Incident #71922. • New steel frame design examples have been added for CSA S16-14 and KBC 2009. • New concrete frame design examples have been added for ACI 318-14, CSA A23.3-14, and KBC 2009.

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		• New shear wall design examples have been added for ACI 318-14, CSA A23.3-14, and KBC 2009.
5	7 July 2016	• Added SAFE design verification examples for slab design, punching shear design, and post-tension design for all codes supported in both SAFE and ETABS.
6	3 March 2017	• Updated benchmark results for Example 14 due to changes reported under Incident #73043.
7	19 June 2017	• Updated benchmarks (number of shear studs) for Composite Beam Design examples AISC-360-05 Example 001 and AISC-360-10 Example 002 due to changes reported under Incident #102064.
8	30 January 2018	• Added Steel Frame Design examples AISC-360-16 Example 001 and AISC-360-16 Example 002 for the addition of the new AISC 360-16 steel frame design code under Incident # 200689. • Added Composite Beam Design examples AISC-360-16 Example 001 and AISC-360-16 Example 002 for the addition of the new AISC 360-16 composite beam design code under Incident # 200897. Documentation for Example AISC 360-10 Example 002 was updated for cosmetic consistency with the new examples, but no results have changed. • Added Composite Beam Design examples AISC DG02 Example 001, AISC DG02 Example 002, AISC DG02 Example 003, and AISC DG02 Example 004 for the new design of steel beams with web openings as implemented under Incident #208550. • Added Composite Beam Design example AISC DG11 2nd Edition Example 001 for the implementation of AISC Design Guide 11 2nd Edition vibration check under Incident #29003. • Removed the Slab Design examples for the AS 3600-2001 and Hong Kong CP 2004 codes, which have been removed from the software under Incident #210114. • Updated the documentation for examples CSA S16-09 Example 002 and CSA S16-14 Example 002 to reflect changes

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Revision Number	**Date**	**Description**
		in hand calculations due to Incidents #201669 and #204405, although these changes did not affect the final design results obtained by the software. • Updated the documentation for example EN 2-2004 Example 002 to reflect changes in design results produced by the software due to Incident #203061. The change was insignificant for the purposes of this verification example.
9	10 June 2019	• Removed steel-frame CSI S16-09, concrete frame CSA A23.3-04, shear wall CSA A23.3-04, and slab CSA A23.3-04 design examples for Incident #234426. • Added composite beam AISC DG31 design examples for Incident #225308. • Added concrete frame AS 3600-2018 design examples for Incident #226394. • Added Shear wall AS 3600-2018 design examples for Incident #236964. • Updated concrete frame HK CP-2004 design examples to HK CP-2013 for incident #217089. • Updated shear wall HK CP-2004 design examples to HK CP-2013 for incident #217089. • Added steel frame KBC 2016 design examples for Incident #226396. • Added concrete frame KBC 2016 design examples for Incident #226395. • Added shear wall KBC 2016 design examples for Incident #236965. • Added concrete frame RCDF 2017 design examples for Incident #226393. • Added shear wall RCDF 2017 design examples for Incident #236966. • Added concrete frame TS 500-2000(R2018) design examples for Incident #226392. • Added shear wall TS 500-2000(R2018) design examples for Incident # 236967.
10	12 December 2019	• Updated the factored moment results in the documentation for concrete slab PT design to match values from the Concrete

ETABS Software Verification Log		
Revision Number	**Date**	**Description**
		Slab Design output tables rather than the analysis tables. The change was insignificant for the purposes of this verification example. This change was done as a part of Ticket #2870.
11	28 September 2020	Initial release of ETABS version 19.0. Updated Composite Beam Verification Examples for minor numerical differences (AISC 360-10, AISC 360-16 and CSA S16-14). Added Composite Column Verification Examples for AISC 360-16. Added ACI 318-19 Concrete Design Verification problems (for slabs, frames and shear walls)
12	June 29, 2021	Added ACI 318- 2019 Example 01 and ACI 318-2019 Example 02 verification problems for concrete frames. Added ACI 318-19 WALL-001 and ACI 318-19 WALL-002 verification problems for shear wall. Added AISC-360-16 Examples 001 through 006 verification problems for Composite Column Design. Added ACI 318-19 RC-PN Example 001, ACI 318-19 RC-PN Example 002, ACI 318-19 PT-SL Example 001, ACI 318-19 RC-SL Example 001 verification problems for Slab Design.
13	November, 2021	Added EC 2-2004 Examples 003 and 004 as well as NTC 2008 Examples 003 and 004 for concrete columns. Modified EC 2-2004 Examples 001 and 002 for concrete columns to include various national annexes.
14	February 2022	Added composite beam Example 001 for CSA S16-2019 per ticket 8195.

ETABS Software Verification Log		
Revision Number	**Date**	**Description**
		Added steel frame design examples 001 and 002 for CSA S16-19 per Ticket 8238. Added concrete frame design examples 001 and 002 for CSA S23.3-19 per ticket 8237.
15	June 2022	Added CSA A23.3-19 RC-PN Example 001, CSA A23.3-19 PT-SL Example 001, CSA A23.3-19 RC-SL Example 001 verification problems for Slab Design.
16	October 2022	Added design examples 001 and 002 for composite column design per EC-4-2004.
17	March 2023	Added design examples 001 through 003 for steel frame design according to AS 4100-2020.
18	December 2023	Added Steel Connection design examples for AISC 360-10 & 16.
19	February 2024	Added Steel Connection, Frame and Composite Beam & Column design examples for AISC 360-22. Added Composite Column Design examples for CSA S16-19.
20	August 2024	Updated the following CBD examples for minor change to fillet radius: AISC DG31 Examples 003 & 004, AISC 360-16 Example 003, BS-5950-90 Example 001, EC-4-2004 Example 001.
21	September 2024	Updated the High Frequency calculations for the AISC-DG11-2ED Ex00l example problem. Added CBD example “IS-11384 Example 001” for the 2022 Indian code.
22	October 2024	Minor correction to Composite Beam problem IS-11384 Example 001
23	November 2024	Added SCI Publication 355 example 001 which deals with composite bending of a beam with web openings.

24	February 2025	Added verification problem AISC 360-22 Wall-001 for composite shear wall design. Update to BS 5950-90 Example 01 to account for new changes where calculations include axial load effects on beam. Added Verification problems CSA S16-24 Ex 001 and 002 for Steel frame design. Added composite column design verification problems for CSA-S16-24 Examples 001 and 002. Added composite beam design verification problem CSA S16-24 Example 001.
25	May 2025	Added AISC DG11 2nd Ed Example 002 for the vibration checks of composite beams.

INTRODUCTION

This manual provides example problems used to test various features and capabilities of the ETABS program. Users should supplement these examples as necessary for verifying their particular application of the software.

METHODOLOGY

A series of test problems, or examples, designed to test the various elements and analysis features of the program were created. For each example, this manual contains a short description of the problem; a list of significant ETABS options tested; and a comparison of key results with theoretical results or results from other computer programs. The comparison of the ETABS results with results obtained from independent sources is provided in tabular form as part of each example.

To validate and verify ETABS results, the test problems were run on a PC platform that was a Dell machine with a Pentium III processor and 512 MB of RAM operating on a Windows XP operating system.

Acceptance Criteria

The comparison of the ETABS validation and verification example results with independent results is typically characterized in one of the following three ways.

Exact: There is no difference between the ETABS results and the independent results within the larger of the accuracy of the typical ETABS output and the accuracy of the independent result.

Acceptable: For force, moment and displacement values, the difference between the ETABS results and the independent results does not exceed five percent (5%). For internal force and stress values, the difference between the ETABS results and the independent results does not exceed ten percent (10%). For experimental values, the difference between the ETABS results and the independent results does not exceed twenty five percent (25%).

Unacceptable: For force, moment and displacement values, the difference between the ETABS results and the independent results exceeds five percent (5%). For internal force and stress values, the difference between the ETABS results and the independent results exceeds ten percent (10%). For experimental values, the difference between the ETABS results and the independent results exceeds twenty five percent (25%).

The percentage difference between results is typically calculated using the following formula:

$$\text{Percent Difference} = 100\left[\frac{\text{ETABS Result}}{\text{Independent Result}} - 1\right]$$

For examples with multiple versions of meshing density of area elements, only the models with the finest meshing density are expected to fall within Exact or Acceptable limits.

Summary of Examples

The example problems addressed plane frame, three-dimensional, and wall structures as well as shear wall and floor objects. The analyses completed included dynamic response spectrum, eigenvalue, nonlinear time history, and static gravity and lateral load.

Other program features tested include treatment of automatic generation of seismic and wind loads, automatic story mass calculation, biaxial friction pendulum and biaxial hysteretic elements, brace and column members with no bending stiffness, column pinned end connections, multiple diaphragms, non-rigid joint offsets on beams and columns, panel zones, point assignments, rigid joint offsets, section properties automatically recovered from the database, uniaxial damper element, uniaxial gap elements, vertical beam span loading and user specified lateral loads and section properties.

Slab examples S01 through S07 verify the accuracy of the elements and the solution algorithms used in ETABS. These examples compare displacements and member internal forces computed by ETABS with known theoretical solutions for various slab support and load conditions.

Slab examples S08 through S014 verify the applicability of ETABS in calculating design moments in slabs by comparing results for practical slab geometries with experimental results and/or results using ACI 318-95 recommendations.

Slab design examples verify the design algorithms used in ETABS for flexural, shear design of beam; flexural and punching shear of reinforced concrete slab; and flexural design and serviceability stress checks of post-tensioned slab by comparing ETABS results with hand calculations.

Analysis: Of the fifteen Analysis problems, eight showed exact agreement while the remaining seven showed acceptable agreement between ETABS and the cited independent sources.

Analysis – Slab: All fourteen slab Analysis problems showed acceptable agreement between ETABS and the cited independent sources.

Design – Steel Frame: All 32 Steel Frame Design problems showed acceptable agreement between ETABS and the cited independent sources.

Design – Concrete Frame: All 40 Concrete Frame Design problems showed acceptable agreement between ETABS and the cited independent sources.

Design – Shear Wall: All 38 of the Shear Wall Design problems showed acceptable agreement between ETABS and the cited independent sources.

Design – Composite Beam: The 18 Composite Beam Design problems showed acceptable agreement between ETABS and the cited independent sources.

Design – Composite Column: The 3 Composite Column Design problems showed acceptable agreement between ETABS and cited independent sources.

Design – Slab: The 39 Slab Design problems showed acceptable agreement between ETABS and cited independent sources.

CONCLUSIONS

ETABS is the latest release of the ETABS series of computer programs. Since development, ETABS has been used widely for structural analysis. The ongoing usage of the program coupled with continuing program upgrades are strong indicators that most program bugs have been identified and corrected.

Additionally, the verification process conducted as described in this document demonstrates that the program features tested are operating reliably and with accuracy consistent with current computer technology capabilities.

Meshing of Area Elements

It is important to adequately mesh area elements to obtain satisfactory results. The art of creating area element models includes determining what constitutes an adequate mesh. In general, meshes should always be two or more elements wide. Rectangular elements give the best results and the aspect ratio should not be excessive. A tighter mesh may be needed in areas where the stress is high or the stress is changing quickly.

When reviewing results, the following process can help determine if the mesh is adequate. Pick a joint in a high stress area that has several different area elements connected to it. Review the stress reported for that joint for each of the area elements. If the stresses are similar, the mesh likely is adequate. Otherwise, additional meshing is required. If you choose to view the stresses graphically when using this process, be sure to turn off the stress averaging feature when displaying the stresses.

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